

A Novel Architecture to Measure the Resonance Frequency of Copper Pipes

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(Max Mustermann)

Dresden, 20 January 2026

This template is for use with pdf $\text{\LaTeX}$ and biber. It has been tested with TeX Live 2024 (tlnet-final 2024).

Abstract

about 1/2 page:

1. Motivation (Why do we care?)
2. Problem statement (What problem are we trying to solve?)
3. Approach (How did we go about it)
4. Results (What's the answer?)
5. Conclusion (What are the implications of the answer?)

The abstract is a miniature version of the thesis. It should be treated as an entirely separate document. Do not assume that a reader who has access to an abstract will also have access to the thesis. Do not assume that a reader who reads the thesis has read the abstract.

Kurzfassung

Gleicher Text (sinngemäß, nicht wörtlich) in Deutsch

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Chapter 1

Guidelines

The following are general guidelines for how to write a thesis, with a special focus on how to use the template we provide. Please read it in full before starting.

This template already includes a large number of LaTeX packages. Make sure you use an up-to-date LaTeX distribution such as *TeX Live*.¹ Familiarize yourself with the available package options of this template (such as the `germanthesis` and `watermark` options).

1.1 Typesetting

You can include many non-ASCII characters (like umlauts `äöü` or accented characters in general `ñçø`) verbatim in your `.tex` code, but make sure to save the file using UTF-8 encoding.

Use quotation marks only for text quoted verbatim from other work “like this” (but consider whether a verbatim quotation is really more appropriate than alternatives). For typesetting quotation marks, consider using the `enquote` command to automatically manage things like changes in language, nested quotations, and the like. Use bold text **very sparingly**, as it is likely to interrupt a reader’s flow of reading. Instead, use the `emph` command to typeset such text in a *somewhat* less intrusive way (while still indicating a change in the tone of voice).

Use paragraphs to break the content into logical groups. Do not use hard line breaks. In the source file, however, line breaks have proven to simplify editing considerably: putting exactly one sentence per line makes it very easy to move content around, it helps quickly identifying overly long sentences, and allows line-based diff tools to work as expected.

¹<https://www.tug.org/texlive/>

Consider using the `textsc` command to make long runs of capital letters slightly less intrusive. Typesetting PCMCIAPEBKAC as `PCMCIAPEBKAC` makes the text stand out much less on the page.

1.2 Cross References

Cross references are an easy way to point a reader to certain parts of the text. For figures, tables, and equations, they are a must. Sometimes, it is also useful to point a reader to other sections, e.g., to Section 1.2, but this should be used sparingly.

1.3 Figures

A picture says more than a thousand words. Use `jpeg` files for photos, use `png` for screenshots, use `pdf` for everything else. Make sure to include all text required to get a (cursory) understanding of the figure in its caption; it is no problem to have a caption spanning five lines. Do not put a second caption in the figure itself. For closely related content, consider using subfigures. Draw figures yourself, but acknowledge all sources. Check font sizes to make sure the figure is neither unreadably small nor overly big: after typesetting the thesis, text in the figure should be the same size as that in the figure caption.

Example: Figures 1.1a and 1.1b show the distribution of the nodes in the sample setup at time $t = 0$, as well as the initial coverage with a sensing radius of 30 m and the communication graph for a communication range of 50 m.

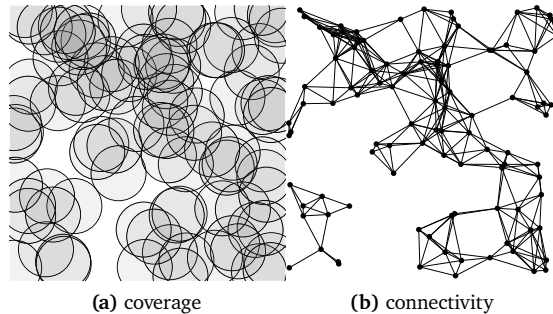


Figure 1.1 – Subfigures showing coverage and connectivity for a sample replication at time $t = 0$

1.4 Tables

Use short tables, e.g., Table 1.1 to give a brief overview of related information.

For presenting result data, a figure is much more appropriate. Multi-line cells can be typeset as shown in Table 1.2.

left aligned	same here	right aligned
1	2	3
4	5	6
7	8	9

Table 1.1 – Short table

Class	application examples	lifetime aspects
Critical, coverage	Forest fire detection, flood detection, nuclear/chemical/biological attack detection, battlefield surveillance, intrusion detection	$c_{ca}/c_{ct}/c_{cb}, c_{ln}, c_{la}, c_{lo}$
Critical, no coverage	Monitoring human physiological data, military monitoring of friendly forces, machine monitoring	$c_{cc}, c_{ln}, c_{la}, c_{lo}$
Noncritical, coverage	Agriculture, smart buildings, habitat monitoring (sensors monitor the inhabitants in a region)	$c_{ac}/c_{tc}/c_{bc}, c_{cc}, c_{sd}$
Noncritical, no coverage	Home automation, habitat monitoring (sensors are attached to animals and monitor their health and social contacts)	c_{cc}, c_{sd}

Table 1.2 – Sensor network applications

1.5 Math

Use *math* mode for typesetting formulas. This will interpret all numbers as quantities and all letters as single-letter variable names, selecting an upright font for numbers and an italic font (and adding some space) for letters. This is ideal for simply typing out equations that only consist of numbers and single-letter variables, e.g., $x = 2ab$. However, this interpretation also means that care is required when typesetting anything else – as it is generally assumed that whitespace between variables means implied multiplication and that italic letters represent variables, while upright letters represent units or descriptive terms.² Consequently:

multi-letter variables require the use of the `\mathit` command to typeset them with the correct font and spacing, so that $x = fiji$ is correctly formatted as $x = \mathit{fiji}$.

descriptive terms should be typeset using the `\mathrm` command, so that $x = y_{direct}$ and its abbreviation $x = y_d$ are correctly formatted as $x = y_{\mathrm{direct}}$ and $x = y_{\mathrm{d}}$, respectively. Note the difference between typesetting “ y_d ” if d represents a quantity (e.g., the index 0) and “ y_d ” if d is a descriptive term (e.g., short for “direct”).

descriptive text requires the use of the `\text` command to typeset it with the correct (non-math) font and spacing, so that “ $x = 2ab$ (for first $a, b \in \mathbb{R}$)” or “ $x = 2ab$ (for first $a, b \in \mathbb{R}$)” are correctly formatted as “ $x = 2ab$ (for first $a, b \in \mathbb{R}$)”.

textual operators require the use of, e.g., the `\mathop` (and `\mathrm`) commands to typeset them with the correct font and spacing, so that $x = z \sin(y)$ is correctly formatted as $x = z \sin(y)$. A more convenient alternative is the use of the `\operatorname` command, resulting in $x = z \sin(y)$. For frequently used operators, consider defining a new operator with the `\DeclareMathOperator` command, resulting in $x = z \sin(y)$.

units should be typeset using the `\SI` command, so that $x = 5\text{m/s}$ is correctly formatted as $x = 5\text{ m/s}$ (see the corresponding section).

Equations are frequently referred to by other papers. Make sure that they are broken up into logical groups and that each of your equations is numbered. Use in-line equations for very simple statements only. To give an example, $\zeta(t) = \min(\zeta_{**}(t))$ is much better written as

$$\zeta(t) = \min(\zeta_{**}(t)), \quad (1.1)$$

²<https://www.nist.gov/pml/special-publication-811/nist-guide-si-chapter-10-more-printing-and-using-symbols-and-numbers>

which is also easier to find.

Equations covering multiple lines should be aligned. Note that the numbering is (and should be) added, independent of whether the equation is actually referenced or not, as in

$$s_{\max} = \max ((t_{i+1} - t_i) : \zeta(t_i) < 1, i \in [0, |T| - 1]), \quad (1.2)$$

$$\psi_s(t) = \begin{cases} \frac{\Delta t_s}{s_{\max}} & \text{if } s_{\max} > 0, \\ 1 & \text{if } s_{\max} = 0, \end{cases} \quad (1.3)$$

$$\zeta_s(t) = \frac{\psi_s - l_s}{c_s - l_s}. \quad (1.4)$$

The *User's Guide for the amsmath Package*³, provided by the American Mathematical Society, has a comprehensive overview of best practices for typesetting mathematical content.

³<https://mirrors.ctan.org/macros/latex/required/amsmath/amsldoc.pdf>

1.6 Units

Numbers with units should be set using the `\SI` command, which enforces consistency throughout the text. For example: “the measurements show that the car was accelerating at 5 m/s^2 until it reached its final speed of 100 km/h . During this it sent 512 bit long packets for a total of 1 MByte/s , more than could be decoded.”

Units without a number can be typeset using `\si`; longer unit-less numbers or ranges can be typeset using the `\num` and `\numrange` commands, respectively. For example: “The number 12 345 678 lies in the range of 10 000 000...20 000 000.” Table 1.3 gives an example of how to typeset numbers and units in tables.

factor		value	unit
M	vehicle mass	1.3250×10^3	kg
g	gravitational constant	9.81	m/s^2
ϑ	road grade	0	°
α		1.1100	g/s
δ		1.9800×10^{-6}	$\text{g s}^2/\text{m}^3$

Table 1.3 – EMIT factors for a category 9 vehicle

1.7 Algorithms

Use algorithms to explain what would be too hard to express in the text body of your thesis.

For example: Based on the periodically transmitted hello messages, the joining node gets information about its physical neighbors and their adjacent nodes. Algorithm 1.1 depicts the handling of hello messages.

Require: Locally stored state of all neighbors in set N
Ensure: Maintain neighbor set N and set virtual address

```

1: Receive neighbor information from node  $N_i$ 
2: if  $N_i \notin N$  then
3:    $N \leftarrow N_i$ 
4: else
5:   Update  $N_i \in N$ 
6: end if
7: if  $P == -1$  AND  $(\text{Time}() - \text{OldTime}) > T_{ps}$  then
8:    $\text{OldTime} \leftarrow \text{Time}()$ 
9:    $\text{SetMyPosition}()$ 
10: end if

```

Algorithm 1.1 – Handle hello messages

1.8 Typesetting Grammars

Grammars can be typeset using the grammar environment.

```

<input> ::= <primary> ';' <input>
| <primary>
|

```

```

<primary> ::= <term> | NUMBER

```

```

<term> ::= <primary> <operator> <primary>

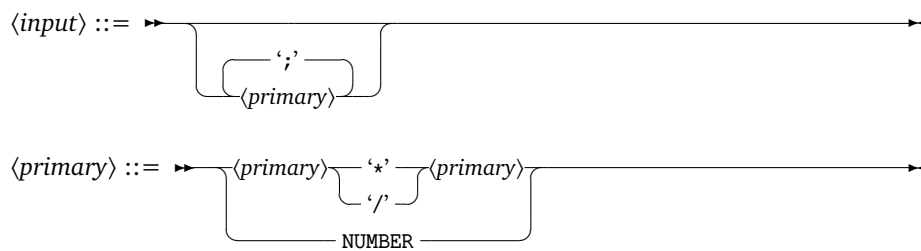
```

```

<operator> ::= '*' | '/'

```

An alternative graphical form is also available. This version is often easier to read, but does not lend itself well to more complex grammars.



1.9 Program Code

Program code should be omitted. What your programs do will already be explained in text form or, if needed, in algorithmic form. If including source code is absolutely necessary, it should be typeset as seen in Listing 1.1.

```
1 APPLICATION("printU", 192, arg)
2 {
3     // Set Priority
4     NutThreadSetPriority(16);
5     // main() loop
6     for (;;) {
7         putchar('U');
8         NutSleep(125);
9     }
10 }
```

Listing 1.1 – Sample application

1.10 References

What follows is just a very quick refresher on how to use references. It is not a guide on scientific writing in general, nor copyright and plagiarism in particular. Please refer to an actual guide on technical writing and scientific practices to make sure you understand how, where, and when to cite.

Simply speaking, proper scientific writing has to deal with two closely related (but not identical) concepts:

a) Copyright

b) Plagiarism

Do not confuse the need for properly citing your sources as something related to copyright. Questions of (a) copyright or the corresponding national equivalent deal with who has the right to reproduce a certain text excerpt, an image, or something similar. Questions of (b) plagiarism deal with who came up with a certain idea or insight, e.g., a certain finding, a certain concept, or a certain way of illustrating a concept. By way of analogy, consider a car: after buying a car you have the right to (a) do whatever you want with it, but you still cannot claim that you (b) invented it. Conversely, properly (b) crediting who invented your neighbor's car does not give you the right to (a) use it. Put yet another way, problem (a) is a legal one: to be allowed to publish a scientific work you (or, rather, your publisher) needs to have permission to reproduce it – or suffer legal consequences like heavy fines. Problem (b) is an academic one: claiming someone else's ideas as one's own is plagiarism; similarly, re-selling old ideas as new ones is self-plagiarism. Both incur heavy penalties like exclusion from schools and professional associations or being blacklisted from publishing with scientific outlets for any number of years.

You will need to address both problems in writing your thesis. Problem (a) can be addressed in two ways: First, by creating original content (that is, text or figures) yourself, which is always preferable as this gives you the freedom to present the content your way. Second, by obtaining a license to reproduce content (e.g., by way of buying a license or adhering to the terms of an existing copyleft license). Problem (b) can be addressed in two ways: First, presenting original ideas and insights (as you will do when presenting own results). Second, by clearly pointing out the (primary) source of an idea. The latter is the topic of this section.

In brief, use references whenever you cite from related work (either directly or indirectly), or when you build on related work (this includes their way of illustrating a particular concept, in text form as well as in the overall design of a figure). Also use references to point a reader to related work. Clearly distinguish between these uses.

Use quotation marks (“...” for direct citations (and direct citations only). Using direct citations where an indirect one would be more appropriate is bad scientific writing. However, when citing indirectly, be sure to actually put the cited idea into your own words: using verbatim text from a source without quotation marks means claiming that you came up with the way of expressing the idea yourself, which is plagiarism.

Make it very clear which part of a statement a reference belongs to. Compare the following three, vastly different uses (where the cited idea appears in **boldface**):

- “**Foo and bar are of equal value. Thus, any can be used.**” [1], [2]
- According to [1] and [2], **foo and bar are equally-valued; from this the authors conclude that any can be used.**

versus

- “**Foo and bar are of equal value**” [1], [2]. Thus, any can be used.
- According to [1] and [2], **foo and bar are equally-valued**. From this it follows that any can be used.

versus

- Foo and **bar** [1], [2] are of equal value. Thus, any can be used.
- Foo and **bar** (detailed in [1] and [2]) are of equal value. Thus, any can be used.

versus

- **Foo** [1] and **bar** [2] are of equal value. Thus, any can be used.
- **Foo** (detailed in [1]) and **bar** (detailed in [2]) are of equal value. Thus, any can be used.

Never typeset a reference after the final full stop of a paragraph (or sentence) and expect your reader to figure out which part of the paragraph is an indirect citation and which part is original (i.e., your own) work. When paraphrasing longer passages of text, use an indirect citation. Make sure to clearly point out when you are finished paraphrasing, like so: *According to Knuth [1], Foo and Bar can be characterized as follows. They are big. They are bright. The authors further argue that one can be substituted for the other. In the following I will go on to prove that this is not true.*

When citing more than a few pages worth of text, point the reader to the specific part you are referring to in your citation, like so: *In recent years, an increasing number of cyclists are switching from air filled tires to cement filled ones [3, Table IV].*

If a figure or a table is closely based on another one, make sure to cite its source, preferably in its caption, like so: *Figure 1 – the relation of ravens and writing desks (based on [4, Figure 42])*. Be aware that, while there is a well-established convention on how to illustrate a verbatim quote of text (by using quotation marks), there is no well-established convention for indicating that an image was copied verbatim. Thus, when citing a figure or table, you must explicitly state whether it was copied verbatim, abstracted, or whether it served as inspiration for your own.

Do not cite URLs Content found there is not peer reviewed and it is likely to change during the lifetime of your work. For pointing a reader to interesting websites, use footnotes – but trust your reader to know how to use a web search engine.

Your text reads nicer if you do not use citations as a substitute for nouns (like this section did). Instead of *The benefits of cement filled tires has been shown by [5]*, consider writing *Hardes and Sommer [5] have shown the benefits of cement filled tires*. The `textcite` command makes this straightforward.

Make sure to read your bibliography section (that is, the typeset list of references) after you are done adding all citations to your text. Does it contain all information needed to uniquely identify to references you used?

Do not trust BibTeX files you find on the web: Digital libraries frequently have their contents wrong, are missing information, or are using different field names than your bibliography style expects (leading to missing information in the typeset bibliography). To give a few examples: Check the authors' list (making sure all authors are listed in the same order and in the same way they are listed in the publication). Check the conference location (it's most likely not "New York, New York"). Check the publisher name (many digital libraries use a field that is not typeset by your bibliography style; have a look at the demo bibliography in this template for how to deal with that). Check the page numbers (many digital libraries put something like "1–5" here despite the paper starting at a later page – or despite it not having any page numbers to begin with). Check the conference name, put its parts in a logical order, and lose the "in proceedings of" (it's not "Mobicom, in proceedings of, 1999 series MobiCom99" but "5th ACM International Conference on Mobile Computing and Networking (MobiCom 1999)". Check articles and capitalization (see, e.g., references [6], [7] for examples).

1.11 Abbreviations

Abbreviations should be explained when first used. Latex helps by providing an `\ac` command that will automatically expand an abbreviation on first use.

first use: Mobile Ad Hoc Networks (MANETs) have been frequently used as examples for the development of Wireless Sensor Network (WSN) applications. Simulations are made more efficient by the use of a Region of Interest (ROI).

second use: MANETs have been frequently used as examples for the development of WSN applications. Simulations are made more efficient by the use of an ROI.

Make sure to explicitly choose either the short `\acs*`, long `\acl*`, or full `\acf*` starred form in headings and captions to avoid unpleasant surprises.

Automatic abbreviations also support ones that only make sense in a foreign language – like that of the German Automobile Association (Allgemeiner Deutscher Automobilclub, ADAC). Some care must be taken when the same abbreviation can refer to multiple things – like Controller Area Network (CAN) and Content Addressable Network (CAN).

In general, use (explained) abbreviations sparingly: Your reader can be assumed to know what a MAC is, what UMTS or GPS do, etc (and, if not, spelling these terms out won't be of much help, anyway). Further, each newly introduced abbreviation increases your reader's cognitive load (you are expecting your reader to be able to recall any of your introduced abbreviations at any later point in the text).

1.12 TODOs and FIXMEs

You can use the `\TODO` command to add short “sticky notes” to your document.

This is what a TODO looks like

This will also trigger generation of a list-of-TODOs at the end of the document. The same goes for the `\FIXME` command.

This is what a FIXME looks like

1.13 Sections

Keep the number of sections and sub-sections low. If two sections start on the same page, you are most likely using too many sections. Each section should include some text (i.e., the heading of section 1 should not be immediately followed by the heading of section 1.1)

Stay clear of *meta* writing, that is, explicitly addressing the way your thesis is laid out: Rather than “this section contains two sub-sections”, say “in the following, I will address two important results: foo and bar”.

Do not use section headings as a replacement for introductory text and a short motivation. It should be easy enough to follow your thesis with all section headings removed.

1.14 About the remainder of this document

The remainder of this document gives a rough outline of how to structure your thesis. It is mainly geared towards Bachelor and Master’s theses, but feel free to adapt the structure to your specific needs. If, however, you are writing a seminar thesis, the present structure is likely wholly inapplicable.

Chapter 2

Introduction

- general motivation for your work, context and goals.
- context: make sure to link where your work fits in
- problem: gap in knowledge, too expensive, too slow, a deficiency, superseded technology
- strategy: the way you will address the problem
- recommended length: 1-2 pages.

Chapter 3

Fundamentals

- describe methods and techniques that build the basis of your work
- include what's needed to understand your work (e.g., techniques, protocols, models, hardware, software, ...)
- exclude what's not (e.g., anything you yourself did, anything your reader can be expected to know, ...)
- review related work(!)
- recommended length: approximately one third of the thesis.

Chapter 4

Developed architecture / System design / Implementation / ...

- describe everything you yourself did (as opposed to the fundamentals chapter, which explains what you built on)
- start with a theoretical approach
- describe the developed system/algorithm/method from a high-level point of view
- go ahead in presenting your developments in more detail
- make sure to instill in your readers a high degree of confidence that your system/algorithm/method is correct – this will likely mean conducting (and reporting on) a validation/verification step
- recommended length: approximately one third of the thesis.

Chapter 5

Evaluation

- measurement setup / results / evaluation / discussion
- whatever you have done, you must comment it, compare it to other systems, evaluate it
- usually, adequate graphs help to show the benefits of your approach
- each result/graph must not only be described, but also discussed (What's the reason for this peak? Why have you observed this effect? What does this tell about your architecture/system/implementation?)
- recommended length: approximately one third of the thesis.

Chapter 6

Conclusion

- summarize again what your paper did, but now emphasize more the results, and comparisons
- write conclusions that can be drawn from the results found and the discussion presented in the paper
- future work (be very brief, explain what, but not much how, do not speculate about results or impact)
- recommended length: one page.

List of Abbreviations

ADAC	German Automobile Association (Allgemeiner Deutscher Automobilclub)
CAN	Controller Area Network, <i>when referring to the bus protocol</i>
CAN	Content Addressable Network, <i>when referring to the distributed hash table</i>
MANET	Mobile Ad Hoc Network
ROI	Region of Interest
WSN	Wireless Sensor Network

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Bibliography

- [1] D. E. Knuth, “The Complexity of Songs,” *Communications of the ACM*, vol. 27, no. 4, Apr. 1984. DOI: 10.1145/358027.358042
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Todo list

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